

Partial Differential Equation (PDE):

A PDE is an equation involving one or more partial derivatives of an (unknown) function, call it u , that depends on two or more variables.

Often time is one variable and one or several variables with space (spatial variables) are involved.

- The order of the highest derivative is called the order of the PDE.
- A PDE is linear if each term in the PDE is of first degree in the unknown function or its partial derivatives. Otherwise we call it nonlinear. Ex: The equation $2u_x + u_y = 0$ is nonlinear.
- We call a linear PDE homogeneous if each of its terms contains either the function or one of its partial derivatives (i.e. if there is no term which doesn't include the function or its derivatives)
Otherwise it is called non homogeneous.

Examples of PDE :

(1) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$: 1-dimensional wave equation

: It can model the vibrating string (and the vibrating membrane)

(2) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$: 1-dimensional heat equation

: It can model temperature in a bar or wire.

(3) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$: 2-dimensional Laplace Equation

: can model Electrostatic Potential.

(4) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$: 2-dimensional Poisson Equation

(5) $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$: 2-dimensional Wave Equation

$$(6) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad : \text{3-dimensional Laplace Equation}$$

- Note: Here c is a positive constant.

t is time, x, y, z are cartesian coordinates and dimension is the number of these coordinates in the equation.

- All the above equations are linear.

The equation (4) (with f not identically zero) is nonhomogeneous. The other equations are homogeneous.

- A solution of a PDE in some region R of the space of independent variables is a function that has all the partial derivatives (appearing in the PDE) in some domain D containing R and satisfies the PDE everywhere in R .

- In general the set of solutions of a PDE can be very large and one

needs some constraints (boundary conditions or initial conditions) to restrict the solution to have physical meaning.

For example, the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

is satisfied by $u = x^2 - y^2$

$$u = e^x \cos(y)$$

$$u = \sin(x) \cosh(y)$$

$$u = \log(x^2 + y^2)$$

Fundamental Theorem on Superposition:

If u_1 and u_2 are solutions of a homogeneous linear PDE in some region R , then

$$u = c_1 u_1 + c_2 u_2$$

with any constants c_1 and c_2 is also a solution of that PDE in the region R .

Ex: Find solutions depending on x and y of
 $u_{xy} + 16u = 0$

- Since no x -derivatives occur, the above is like $u'' + 16u = 0$

The general solution is $u = A \cos(4y) + B \sin(4y)$

Here A and B may be functions of x

and so,

$$u = A(x) \cos(4y) + B(x) \sin(4y)$$

Ex: Find solutions depending on x and y
of $u_{xy} = u_x$

- Setting $v = u_x$ we get

$$v_y = v \Rightarrow \frac{dv}{v} = dy$$

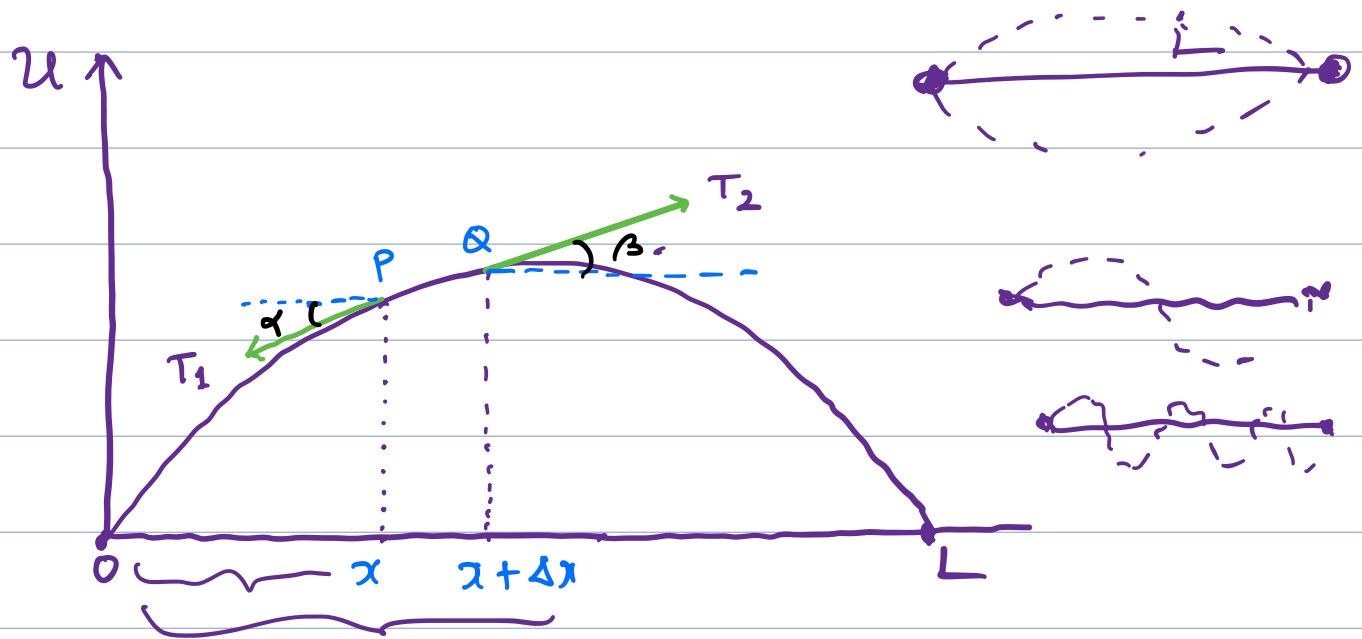
$$\Rightarrow \ln|v| = y + c_1(x) \Rightarrow v = c_2(x) e^y$$

Integrating with respect to x ,

$$u = \int c_2(x) e^y dx = f(x) e^y + g(y)$$

$$\Rightarrow u = f(x) e^y + g(y) \quad \text{where } f(x) = \int c_2(x) dx$$

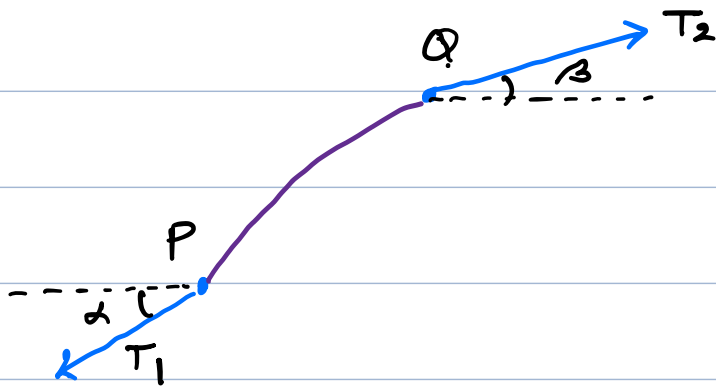
Vibrating String : Wave Equation



Physical assumptions:

- The mass of the string per unit length is constant ("homogeneous string"). The string is perfectly elastic and does not offer any resistance to bending.
- The tension caused by stretching the string before fastening it at the ends is so large that the action of the gravitational force on the string (trying to pull the string down a little) can be neglected.
- The string performs small transverse

motions in a vertical plane, that is, every particle of the string moves strictly vertically and so the deflection and the slope at every point of the string always remain small in absolute value.



Since the string offers no resistance to bending, the tension is tangential to the curve at each point.

Let T_1 and T_2 be tension at the points P and Q.

Since the points move vertically (and not horizontally), the horizontal tension must cancel at every point.

$$\text{Horizontally} \quad T_2 \cos \beta - T_1 \cos \alpha = 0$$

$$\Rightarrow T_2 \cos \beta = T_1 \cos \alpha = T \text{ (constant)}$$

- Vertically the difference of the forces translates into an acceleration.

$$\text{So } T_2 \sin \beta - T_1 \sin \alpha = (\ell \Delta x) u_{tt}$$

where ℓ = mass density of the string

Δx = distance between $P = x$ and $Q = x + \Delta x$

$$\text{Therefore } \frac{T_2 \sin \beta}{T} - \frac{T_1 \sin \alpha}{T} = \frac{(\ell \Delta x)}{T} u_{tt}$$

$$\Rightarrow \frac{T_2 \sin \beta}{T_2 \cos \beta L} - \frac{T_1 \sin \alpha}{T_1 \cos \alpha} = \frac{(\ell \Delta x)}{T} u_{tt}$$

$$\Rightarrow \frac{\tan \beta - \tan \alpha}{\Delta x} = \frac{\ell}{T} u_{tt}$$

$$\Rightarrow \frac{\left(\frac{\partial u}{\partial x}\right)\big|_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)\big|_x}{\Delta x} = \frac{\ell}{T} u_{tt}$$

Taking limit as $\Delta x \rightarrow 0$

$$\text{we have } u_{xx} = \frac{\ell}{T} u_{tt}$$

$$\Rightarrow u_{tt} = \frac{T}{\ell} u_{xx}$$

$$\Rightarrow \boxed{u_{tt} = c^2 u_{xx}}$$

This is the 1D wave equation and c is the propagation speed.

- Thus the model of the vibrating string consists of the 1D wave equation

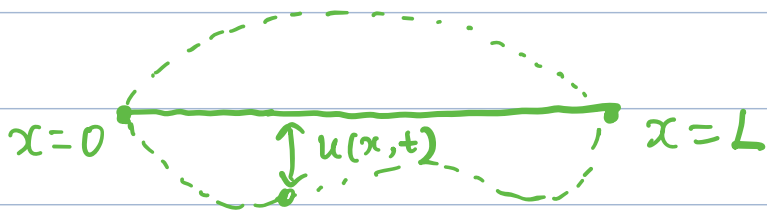
$$u_{tt} = c^2 u_{xx}$$

with some boundary conditions

$$u(0, t) = 0, \quad u(L, t) = 0 \text{ for all } t \geq 0$$

and some initial conditions on the initial shape and velocity of the string

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x) \text{ for } 0 \leq x \leq L$$



The solution has three steps.

- (1) Separating variables
- (2) Satisfying the boundary conditions
- (3) Satisfying the initial conditions.
(using Fourier series)

Step 1: Let us look for a solution of the form:

$$u(x,t) = F(x)G(t)$$

$$\text{Then } u_{tt} = FG_{tt}, \quad u_{xx} = GF_{xx}$$

$$\text{So, the PDE becomes } FG_{tt} = c^2 GF_{xx}$$

$$\Rightarrow \frac{1}{c^2} \frac{G_{tt}}{G} = \frac{F_{xx}}{F}$$

The left hand side depends only on t , while the right hand side depends only on x .

$$\text{Hence } \frac{1}{c^2} \frac{G_{tt}}{G} = \frac{F_{xx}}{F} = k \text{ (constant)}$$

$$\Rightarrow \left. \begin{aligned} F_{xx} - kF &= 0 \\ G_{tt} - c^2 k G &= 0 \end{aligned} \right\}$$

Step 2:

The boundary conditions are

$$u(0,t) = F(0)G(t) = 0$$

$$u(L,t) = F(L)G(t) = 0$$

Now $G(t)$ can't be zero because it will give a solution $u=0$ which is of

no interest.

$$\text{Hence } F(0) = F(L) = 0$$

Now consider the ODE for F :

$$F_{xx} - kF = 0$$

- If $k=0$, then the general solution is

$$F = ax + b$$

and two boundary conditions will make $a=0$, $b=0$ which is again of no interest.

- If $k = \mu^2 > 0$, then the general solution is : $F = c_1 e^{\mu x} + c_2 e^{-\mu x}$

The two boundary conditions will make $c_1 = 0$, $c_2 = 0$ which is of no interest.

- If $k = -\mu^2 < 0$ then the general solution is $F = c_1 \cos(\mu x) + c_2 \sin(\mu x)$.

Using the two boundary conditions

$$0 = c_1 \times 1 + c_2 \times 0 \Rightarrow c_1 = 0$$

$$\text{and } 0 = c_2 \sin(\mu L)$$

$$\Rightarrow \sin(\mu L) = 0 \Rightarrow \mu L = n\pi$$

$$\Rightarrow \mu = \frac{n\pi}{L}$$

Thus there are infinitely many solutions of the form $F(x) = F_n(x) = \sin\left(\frac{n\pi}{L}x\right)$

$$\text{We now solve } G_{tt} - c^2 k G = 0$$

$$\text{for } k = -\left(\frac{n\pi}{L}\right)^2$$

$$\text{Define } \lambda_n = c\mu = \frac{c\pi}{L}n$$

$$\text{Then } G_{tt} + \lambda_n^2 G = 0$$

The general solution is

$$G(t) = a_n \cos(\lambda_n t) + b_n \sin(\lambda_n t)$$

Hence the solutions of the wave equation satisfying the boundary conditions can be written as:

$$u_n(x, t) = \left[a_n \cos(\lambda_n t) + b_n \sin(\lambda_n t) \right] \sin\left(\frac{n\pi}{L}x\right)$$

for $n = 1, 2, \dots$

These functions are called eigen functions and $\lambda_n = \frac{c\pi}{L}n$ are called

eigen values of the vibrating string.

The set $\{\lambda_1, \lambda_2, \dots\}$ is called the spectrum.

$$\text{So, } u_n(x, t) = \left[a_n \cos\left(\frac{c\pi n}{L}t\right) + b_n \sin\left(\frac{c\pi n}{L}t\right) \right] \sin\left(\frac{n\pi}{L}x\right) \\ \text{for } n=1, 2, \dots$$

- Each u_n represents a harmonic motion having the frequency $\frac{\lambda_n}{2\pi} = \frac{c\pi n}{2\pi} = \frac{cn}{2L}$ (cycles/unit time)

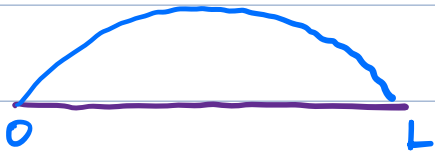
This motion is called n th normal mode of the string.

The first normal mode is called the fundamental mode ($n=1$) and the others are called overtones.

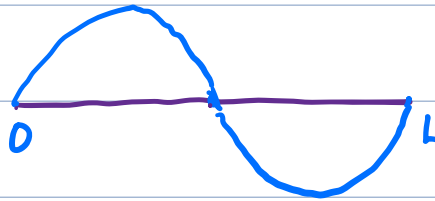
$$\text{Now } \sin\left(\frac{n\pi x}{L}\right) = 0 \text{ at } x = \frac{L}{n}, \frac{2L}{n}, \dots, \frac{(n-1)L}{n},$$

the n th normal mode has $(n-1)$ nodes,

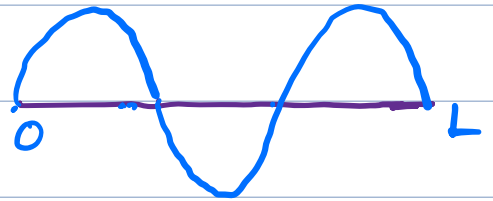
ie. the points of the string that do not move (in addition to the fixed end points)



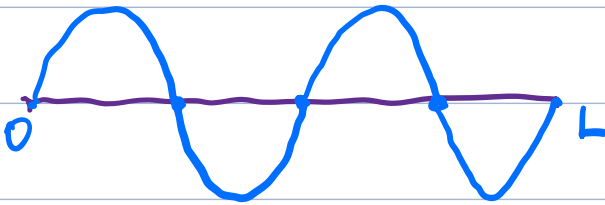
$n=1$



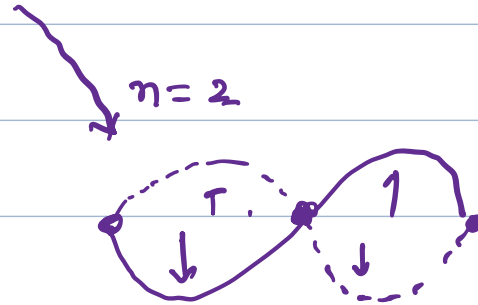
$n=2$



$n=3$



$n=4$



Note that $c = \sqrt{\frac{T}{\rho}}$ so that tuning an

instrument amounts to changing T and ultimately c . The other two variables to control are l and L .

Step 3:

The general solution of the vibrating string is:

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t)$$

$$= \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L} x\right) \left(a_n \cos(\lambda_n t) + b_n \sin(\lambda_n t)\right)$$

For the initial shape condition we have

$$u(x, 0) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi}{L} x\right) = f(x)$$

If we expand $f(x)$ in its fourier series assuming we make an odd extension of it and make it of period $2L$, then f can be expressed as

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2}{L} \int_0^L f(v) \sin\left(\frac{n\pi}{L} v\right) dv \right) \sin\left(\frac{n\pi}{L} x\right)$$

Comparing,

$$a_n = \frac{2}{L} \int_0^L f(v) \sin\left(\frac{n\pi}{L} v\right) dv$$

Now for initial speed, we take the derivative of the general solution:

$$u_t(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L} x\right) \left(-a_n \lambda_n \sin(\lambda_n t) + b_n \lambda_n \cos(\lambda_n t) \right)$$

Now $u_t(x, 0) = \sum_{n=1}^{\infty} b_n \lambda_n \sin\left(\frac{n\pi}{L} x\right) = g(x)$

If we expand $g(x)$ in its fourier series assuming we make an odd extension of it and make it of period $2L$, then g can be expressed as

$$g(x) = \sum_{n=1}^{\infty} \left\{ \frac{2}{L} \int_0^L g(v) \sin\left(\frac{n\pi}{L} v\right) dv \right\} \sin\left(\frac{n\pi}{L} x\right)$$

Comparing,

$$b_n = \frac{2}{\lambda_n L} \int_0^L g(v) \sin\left(\frac{n\pi}{L} v\right) dv$$

Finally the particular solution is

$$u(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L} x\right) \left(a_n \cos(\lambda_n t) + b_n \sin(\lambda_n t) \right)$$

$$\text{where } a_n = \frac{2}{L} \int_0^L f(v) \sin\left(\frac{n\pi}{L} v\right) dv$$

$$b_n = \frac{2}{\lambda_n L} \int_0^L g(v) \sin\left(\frac{n\pi}{L} v\right) dv$$

$$\lambda_n = c\mu = \frac{c\pi}{L} n$$

• We may reformulate the solution as:

$$u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi}{L} x\right) \cos\left(\frac{c\pi}{L} n t\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right) \sin\left(\frac{c\pi}{L} n t\right)$$

$$= \sum_{n=1}^{\infty} \frac{a_n}{2} \left[\sin\left(\frac{n\pi}{L} (x - ct)\right) + \sin\left(\frac{n\pi}{L} (x + ct)\right) \right] + \sum_{n=1}^{\infty} \frac{b_n}{2} \left[\cos\left(\frac{n\pi}{L} (x - ct)\right) - \cos\left(\frac{n\pi}{L} (x + ct)\right) \right]$$

$$= \sum_{n=1}^{\infty} \left[\frac{a_n}{2} \sin\left(\frac{n\pi}{L}(x-ct)\right) + \frac{b_n}{2} \cos\left(\frac{n\pi}{L}(x-ct)\right) \right] \\ + \sum_{n=1}^{\infty} \left[\frac{a_n}{2} \sin\left(\frac{n\pi}{L}(x+ct)\right) - \frac{b_n}{2} \cos\left(\frac{n\pi}{L}(x+ct)\right) \right]$$

- If we consider the case where the initial velocity $g(x) \equiv 0$, then $b_n = 0 \quad \forall n$

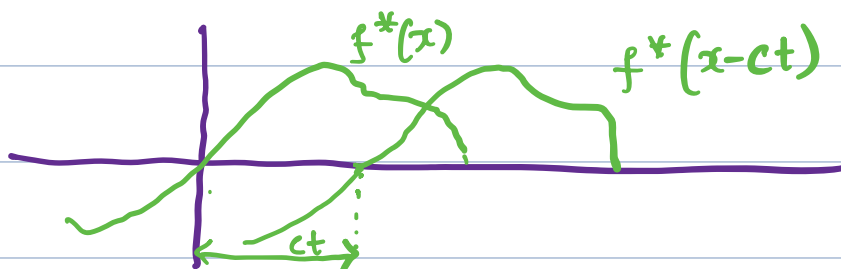
and so the solution $u(x, t)$ reduces to

$$u(x, t) = \sum_{n=1}^{\infty} \frac{a_n}{2} \sin\left(\frac{n\pi}{L}(x-ct)\right) \\ + \sum_{n=1}^{\infty} \frac{a_n}{2} \sin\left(\frac{n\pi}{L}(x+ct)\right)$$

Thus $u(x, t) = \frac{1}{2} \left[f^*(x-ct) + f^*(x+ct) \right]$

$$\left(f^*(\xi) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi}{L}\xi\right) \right)$$

So, u is sum of two travelling waves.



Ex:

$$f(x) = \begin{cases} \frac{2k}{L} x & \text{for } 0 < x < \frac{L}{2} \\ \frac{2k}{L} (L-x) & \text{for } \frac{L}{2} < x < L \end{cases}$$

and $g(x) = 0$

Therefore $b_n = 0$ for all n .

$f(x)$ can be expanded (Half Range Expansion)

Then

$$u(x, t) = \frac{8k}{\pi^2} \left(\frac{1}{1^2} \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{c\pi t}{L}\right) - \frac{1}{3^2} \sin\left(\frac{3\pi x}{L}\right) \cos\left(\frac{3c\pi t}{L}\right) + \dots \right)$$

(• Please see the graphs in the book)

D'Alembert's Solution of the Wave Equation:

The wave equation is $u_{tt} = c^2 u_{xx}$ ($c^2 = \frac{T}{\rho}$)

Let us introduce the variables

$$v = x + ct, \quad w = x - ct$$

$$\text{Then } u_x = u_v v_x + u_w w_x = u_v(1) + u_w(1)$$

$$\Rightarrow u_x = u_v + u_w$$

$$\begin{aligned} u_{xx} &= (u_v + u_w)_x = (u_v + u_w)_v v_x + (u_v + u_w)_w w_x \\ &= (u_{vv} + u_{vw})v_x + (u_{vw} + u_{ww})w_x \\ &= u_{vv} + 2u_{vw} + u_{ww} \end{aligned}$$

$$\text{Now } u_t = u_v v_t + u_w w_t = u_v(c) + u_w(-c)$$

$$\Rightarrow u_t = c(u_v - u_w)$$

$$\begin{aligned} u_{tt} &= c(u_v - u_w)_t = c[(u_v - u_w)_v v_t + (u_v - u_w)_w w_t] \\ &= c[(u_{vv} - u_{vw})c + (u_{vw} - u_{ww})(-c)] \\ &= c^2[u_{vv} - 2u_{vw} + u_{ww}] \end{aligned}$$

$$u_{tt} = c^2 u_{xx}$$

Hence the PDE becomes

$$c^2[u_{vv} - 2u_{vw} + u_{ww}] = c^2[u_{vv} + 2u_{vw} + u_{ww}]$$

$$\Rightarrow -u_{\omega v} = u_{\omega v} \Rightarrow u_{\omega v} = 0$$

Integrating with respect to v we get
$$u_{\omega} = h(\omega)$$

Integrating with respect to ω we get

$$u = \int h(\omega) d\omega = \psi(\omega) + \phi(v)$$

In terms of x and t we have

$$u(x, t) = \phi(x+ct) + \psi(x-ct)$$

where ϕ and ψ are two (possibly different) travelling waves.

This is known as d'Alembert's solution of the wave equation.

Initial Conditions:

$$u(x, t) = \phi(x+ct) + \psi(x-ct)$$

We now impose the initial conditions

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

$$\text{Now } u_t(x, t) = c\phi'(x+ct) - c\psi'(x-ct)$$

$$u(x, 0) = \phi(x) + \psi(x) = f(x)$$

$$u_t(x, 0) = c\phi'(x) - c\psi'(x) = g(x)$$

Then
$$\int_{x_0}^x \phi'(s) ds - \int_{x_0}^x \psi'(s) ds = \frac{1}{c} \int_{x_0}^x g(s) ds$$

$$\Rightarrow \phi(x) - \phi(x_0) - \psi(x) + \psi(x_0) = \frac{1}{c} \int_{x_0}^x g(s) ds$$

$$\Rightarrow \phi(x) - \psi(x) = \phi(x_0) - \psi(x_0) + \frac{1}{c} \int_{x_0}^x g(s) ds$$

Adding this to the equation $\phi(x) + \psi(x) = f(x)$ we get,

$$2\phi(x) = [\phi(x_0) - \psi(x_0)] + f(x) + \frac{1}{c} \int_{x_0}^x g(s) ds$$

$$\Rightarrow 2\phi(x) = k(x_0) + f(x) + \frac{1}{c} \int_{x_0}^x g(s) ds$$

$$\text{where } k(x_0) = \phi(x_0) - \psi(x_0)$$

$$\text{Therefore } \phi(x) = \frac{1}{2} \left[k(x_0) + f(x) + \frac{1}{c} \int_{x_0}^x g(s) ds \right]$$

Similarly subtracting we get

$$2\psi(x) = -k(x_0) + f(x) - \frac{1}{c} \int_{x_0}^x g(s) ds$$

$$\Rightarrow \psi(x) = \frac{1}{2} \left[-k(x_0) + f(x) - \frac{1}{c} \int_{x_0}^x g(s) ds \right]$$

Now the solution of the wave equation is:

$$\begin{aligned}u(x,t) &= \phi(x+ct) + \psi(x-ct) \\&= \frac{1}{2} \left[k(x_0) + f(x+ct) + \frac{1}{c} \int_{x_0}^{x+ct} g(s) ds \right] \\&\quad + \frac{1}{2} \left[-k(x_0) + f(x-ct) - \frac{1}{c} \int_{x_0}^{x-ct} g(s) ds \right]\end{aligned}$$

$$u(x,t) = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

If the initial velocity is zero, this reduces to

$$u(x,t) = \frac{1}{2} [f(x+ct) + f(x-ct)]$$